Methodology

1. Hardware and Software Requirement

a) Hardware:

Laptop/PC with minimum 2 GB RAM and 250 GB storage

Internet connection for dataset access and tool usage

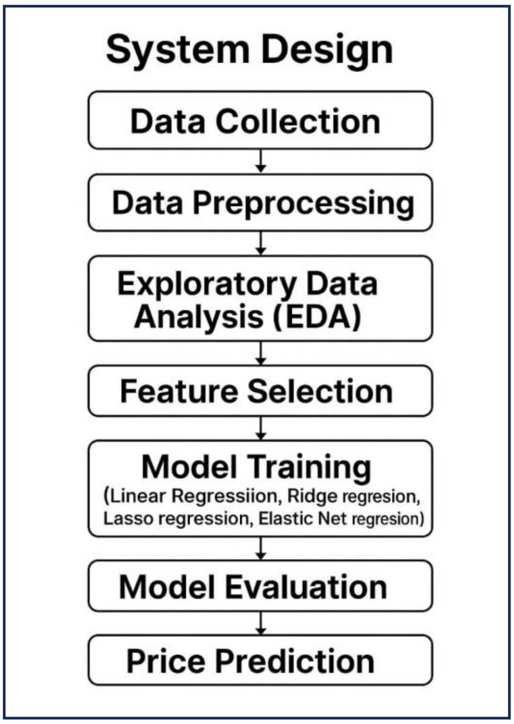
b)Software:

Python (Jupyter Notebook or Google Colab)

Libraries: pandas, numpy, matplotlib, seaborn, scikit-learn

2. System Design (Block Diagram)

You can design a block diagram with the following structure:



3. Algorithm

1. Load and clean the dataset (handle null values, remove duplicates, correct data types)

2. Perform EDA to understand data distributions and relationships

3. Encode categorical variables (cut, color, clarity)

4. Scale numerical features (carat, depth, etc.)

5. Split the data into training and testing sets

6. Train models (e.g., Linear regression, ridge regression, lasso regression, Elastic Net regression)

7. Evaluate the models using metrics like RMSE, R^2 Score

8. Use the best-performing model to predict diamond prices

4. Exploratory Data Analysis and Dataset Visualization

a) Dataset Used:

Diamonds dataset from Kaggle

Features include: carat, cut, color, clarity, depth, table, x, y, z, and price

b) Visualization Tools:

Matplotlib and Seaborn used for plots such as:

Price vs Carat

Price distribution by Cut and Clarity

Correlation Heatmap